

Finite-dimensional fluctuations with thermal noise

HOOF-RIESZ, Rounds 018–019 18 September 2026

Status. THM–040 and the entire THM–041 conjunction pass fresh whole-statement reconstructions AUD–055/057 and separate hostile reviews AUD–056/058. Exact source gates are matched; these finite-dimensional assertions are accepted within their stated scope. The complete mathematical arguments, exact source qualifications, exposure records and sealed diagnostics are the separate Round 018/019 Markdown memoranda and audits. This document is a synthesis, not a substitute for those proofs.

Model and scope. Fix $d \geq 3$, $0 < s \leq d - 2$ and finite $T \geq 0$. On the mass-one torus, initial particles are iid Haar and independent of their independent Brownian drivers. The actual noncolliding singular dynamics is

$$dX_i = \frac{1}{N} \sum_{\ell \neq i} K(X_i - X_\ell) dt + \sqrt{2\nu} dW_i, \quad K = -\nabla g, \quad \hat{g}(k) = c_{d,s} |k|^{s-d} \quad (k \neq 0),$$

where $\hat{g}(0) = 0$ and $c_{d,s} = \pi^{s-d/2} \Gamma((d-s)/2) / \Gamma(s/2)$ for the Fourier convention $e^{2\pi i k \cdot x}$. Set $b(0) = 1$, $b(\nu) = \min(1, \nu^{-1})$ for $\nu > 0$, and $\rho_N = \eta_N - dx$. Thus $b \leq 1$ and $\nu b \leq 1$ for every finite diffusivity. Common translations and pathwise uniqueness preserve the initial translation invariance. Every one-body marginal is exactly Haar, so this centering equals exact first-marginal centering; higher marginals remain correlated.

For the law statements add $s < d/2$ and fix a finite list of deterministic times and real smooth tests $(t_j, h_j)_{j=1}^m$. Let Q_t^ν preserve constants and have nonzero multiplier $e^{-tL_k(\nu)}$, with

$$a_k = 4\pi^2 |k|^2, \quad D_k = 4\pi^2 c_{d,s} |k|^{s+2-d}, \quad L_k(\nu) = D_k + \nu a_k > 0.$$

The covariance is the sum of two Gram matrices:

$$\begin{aligned} C_{ij}(\nu) &= b \int (Q_{t_i}^\nu h_i - \int h_i) (Q_{t_j}^\nu h_j - \int h_j) dx \\ &\quad + 2\nu b \int_0^{t_i \wedge t_j} \int \nabla Q_{t_i-r}^\nu h_i \cdot \nabla Q_{t_j-r}^\nu h_j dx dr. \end{aligned}$$

Equivalently, the exact modal weights are

$$C_{ij}(\nu) = b \sum_{k \neq 0} \widehat{h_i}(k) \overline{\widehat{h_j}(k)} \left[\frac{D_k}{L_k} e^{-(t_i+t_j)L_k} + \frac{\nu a_k}{L_k} e^{-|t_i-t_j|L_k} \right].$$

The initial component has time-sum damping; the thermal component also contains the time difference. Constants, zero/repeated times and singular covariance are included.

The two accepted assertions. For a bounded sequence $\nu_N \rightarrow \bar{\nu} \geq 0$, define the finite vector by

$$Z_{N,j} = \sqrt{Nb(\nu_N)} \rho_N(t_j)[h_j].$$

THM–040 states its joint convergence to the centered Gaussian with covariance $C(\bar{\nu})$. No diffusivity convergence rate is assumed. THM–041 first extends the actual source estimate uniformly over *every finite* $\nu \geq 0$, then states

$$\sup_{0 \leq \nu < \infty} d_{\text{BL}}(\mathcal{L}(Z_{N,\nu}), \mathcal{N}_m(0, C(\nu))) \rightarrow 0,$$

where the distance is the supremum over real functions of sup norm at most one and Lipschitz constant at most one. The supremum is for a fixed finite time/test list. Infinity is never a coefficient in the particle equation.

Source uniformity and the singular identity. The original statistic is

$$P_N[J] = \frac{1}{2N^2} \sum_{i \neq \ell} J(X_i, X_\ell) - \eta_N \left[\int J(\cdot, y) dy \right] + \frac{1}{2} \iint J.$$

For $J_f(x, y) = K(x - y) \cdot (\nabla f(x) - \nabla f(y))$ off the diagonal, its row is $-D * f$ and its scalar contraction is zero, where $D = \text{div } K$ is the full finite measure. At Coulomb, $D = (d-2)|\mathbb{S}^{d-1}|(\delta_0 - dx)$; both atom and compensation remain.

At each fixed positive finite ν , smooth free-energy decrease, nonnegative entropy and fixed- N same-noise heat passage followed by shifted Fatou give $\mathbb{E}_\nu H_N(t) \leq 0$, with $H_N = N^{-1} \sum_{i < \ell} g(X_i - X_\ell)$. At zero noise use deterministic energy decrease. This upper bound is independent of all finite ν , although constants used to construct each individual smooth law may depend on it.

The full Round 016 proof retains the positive heat weight $(1 - e^{-u/r})^{d+2}$ and writes $g = g_r + Q_r - c_r$, with $Q_r \geq 0$. Its exact retained/omitted energy identity, logarithmic Fourier-symbol bound and commutator estimate use only the preceding energy sign and spatial seminorms of the test. Every such seminorm of $Q_{T-t}^\nu h$ is bounded uniformly in all finite ν . The positive omitted heat kernel controls the absolute source and all original contractions. Choosing $r = N^{-2/d}$ gives

$$\sup_{\nu < \infty, t \leq T} \mathbb{E}_\nu |P_N[J_{Q_{T-t}^\nu h}]| \leq C_h N^{s/d-1}.$$

No uniform time-derivative bound as $\nu \rightarrow \infty$ is needed. The original singular observable is integrable by the actual pair-energy moment. The bounded-noise theorem card alone does not prove this extension; the complete proof constants are checked directly.

Stopped smooth one-body Itô calculus, exact ordered-pair symmetrization and removal of collision stops give

$$Z_{N,\nu} = A_{N,\nu} + M_{N,\nu} + R_{N,\nu}, \quad \sup_{\nu < \infty} \mathbb{E} |R_{N,\nu}| \leq C N^{s/d-1/2}.$$

Here $A_j = \sqrt{b/N} \sum_i (Q_{t_j}^\nu h_j(X_i(0)) - \int h_j)$ and

$$M_j = \sqrt{2\nu b/N} \sum_i \int_0^{t_j} \nabla Q_{t_j-r}^\nu h_j(X_i(r)) \cdot dW_i(r).$$

Bounded gradients pass the stochastic integrals in L^2 ; the actual source passes in L^1 . No source-square estimate or singular self value is asserted.

The joint probability argument. Positive heat retention and the actual energy sign give $\mathbb{E} |\hat{\rho}_N(k)|^2 \leq \min(1, C N^{-(1-s/d)} |k|^{d-s})$ uniformly in all finite ν . Absolute Fourier summation therefore controls the first absolute moment of every smooth gradient-product bracket test. For a fixed real vector u , let $M = u \cdot M_{N,\nu}$, $q = \langle M \rangle_T$ and let v be its deterministic Haar bracket. Then

$$\mathbb{E} |q - v| \leq C_u N^{-(1-s/d)/2}, \quad 0 \leq q, v \leq K_u,$$

uniformly in ν , because $\nu b \leq 1$. The complex exponential $\exp(iM_t + \langle M \rangle_t/2)$ is a true martingale with conditional expectation one given the initial sigma field. Multiplying by $e^{iu \cdot A}$ and replacing its bounded compensator by v costs at most $C_u \mathbb{E} |q - v|$. This proves the needed asymptotic factorization without finite- N independence.

The initial bounded iid triangular summands give a uniform characteristic-function error $O_u(N^{-1/2})$. Adding the source error proves compact-uniform characteristic convergence to $e^{-u^T C(\nu)u/2}$. Gaussian convolution, uniform second-moment tails for $A + M$, and the L^1 source remainder prove the full bounded-Lipschitz supremum. Removing the smoothing by a Lipschitz coupling includes degenerate covariances.

Consequences and remaining scope. For finite $\nu_N \rightarrow \bar{\nu}$, dominated Fourier summation gives covariance continuity, and positive semidefinite square-root coupling identifies the Gaussian limit. At $\bar{\nu} = 0$ the thermal part vanishes. If $\nu_N \rightarrow \infty$, then $\text{tr} C(\nu_N) \leq b(\nu_N) \sum_j \|h_j - \int h_j\|_2^2 \rightarrow 0$, so the actual finite vector converges to zero in probability without a growth-rate restriction. The strict threshold $s < d/2$ is needed for the scaled source; equality supplies no decay. No continuous-path, distribution-valued, growing-test, larger-exponent, logarithmic, inhomogeneous or higher-hierarchy theorem is asserted here. THM-042 separately asks for bounded-noise continuous-path convergence. The three original conditions remain distinct:

$$\beta_N N^{2s/d-1} \rightarrow 0, \quad \lambda_N := \beta_N N^{s/d-1} \rightarrow 0, \quad \lambda_N \rightarrow \lambda \in (0, \infty).$$

The restricted homogeneous results do not equate these conditions or resolve every campaign flagship.